Lecture Notes

on Eigenvalues and Eigenvectors

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1 Determinant

The **determinant** is defined only for **square matrices**. For a 2×2 matrix

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix},$$

the determinant is denoted by $\det(A)$ or $|A|$, and is given by

$$\boxed{\det(A) = a_{11}a_{22} - a_{12}a_{21}.}$$

Example

$$\det\begin{pmatrix} 3 & 7 \\ 2 & 5 \end{pmatrix} = 3\times 5 - 2\times 7 = 15 - 14 = 1.$$

1.1 Determinant of a Diagonal Matrix

The determinant of a diagonal, upper triangular, or lower triangular matrix is the **product of its diagonal entries**.

$$A = \begin{pmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{pmatrix} \quad \text{(Diagonal Matrix)}$$

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{pmatrix} \quad \text{(Upper Triangular Matrix)}$$

$$A = \begin{pmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ a_{21} & a_{22} & 0 & \cdots & 0 \\ a_{31} & a_{32} & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{pmatrix} \quad \text{(Lower Triangular Matrix)}$$

In all cases,

$$\boxed{\det(A) = a_{11}a_{22}a_{33}\cdots a_{nn}.}$$

2 Eigenvalues

2.1 Definition

Let A be an $n \times n$ matrix. A **scalar** $\lambda \in \mathbb{R}$ (or $\mathbb{C}$- excluded from your syllabus) is called an **eigenvalue** of A if there exists a nonzero vector $\vec{x}$ such that

$$A\vec{x} = \lambda\vec{x}.$$

$I\vec{x} = \vec{x}$

The corresponding vector $\vec{x}$ is called an **eigenvector** of A associated with the eigenvalue λ.

2.2 Derivation (Characteristic Equation)

Rewriting $A\vec{x} = \lambda\vec{x}$ gives $\Rightarrow A\vec{x} - \lambda\vec{x} = \vec{0}$ $\Rightarrow A\vec{x} - \lambda I\vec{x} = \vec{0}$

$$(A - \lambda I)\vec{x} = \vec{0}.$$

(I - identity matrix, $n \times n$)

For a nonzero solution $\vec{x}$ to exist, the matrix $(A - \lambda I)$ must be **singular** (i.e., det should be zero). Hence,

$$\boxed{\det(A - \lambda I) = 0}$$

If $\det(A) \neq 0$ then A is non-singular

is called the **characteristic equation** of A. The polynomial $\det(A - \lambda I)$ is the **characteristic polynomial**, and its roots are the eigenvalues of A.

Note: The largest absolute value among all eigenvalues is known as the dominant eigenvalue.

$ax = 0$
$\Rightarrow a = 0$
or
$x = 0$

2.3 Example: 2×2 Matrix

Let

$$A = \begin{pmatrix} 7 & 0 \\ -2 & 3 \end{pmatrix}.$$

$(A - \lambda I) = \begin{pmatrix} 7 & 0 \\ -2 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 7 & 0 \\ -2 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$

$= \begin{pmatrix} 7-\lambda & 0 \\ -2 & 3-\lambda \end{pmatrix}$

$\therefore |A - \lambda I| = \begin{vmatrix} 7-\lambda & 0 \\ -2 & 3-\lambda \end{vmatrix}$

$= (7-\lambda)(3-\lambda) - (-2 \times 0)$

$= (7-\lambda)(3-\lambda)$

$\therefore |A - \lambda I| = 0$

$\Rightarrow (7-\lambda)(3-\lambda) = 0.$

$\Rightarrow \lambda = 7, 3.$

Let

$$A = \begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix}.$$

$|A - \lambda I| = 0$

$\Rightarrow \begin{vmatrix} 2-\lambda & 3 \\ 2 & 1-\lambda \end{vmatrix} = 0.$

$\Rightarrow (2-\lambda)(1-\lambda) - 2 \times 3 = 0.$

$$\Rightarrow (2 - 2\lambda - \lambda + \lambda^2) - 6 = 0.$$

$$\boxed{(2-\lambda)(1-\lambda) - 2\times 3 = 0}$$

$$\Rightarrow \underbrace{\lambda^2}_{a=1} \underbrace{-3\lambda}_{b=-3} \underbrace{-4}_{c=-4} = 0.$$

$$ax^2 + bx + c = 0.$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow \lambda = \frac{3 \pm \sqrt{(-3)^2 - 4\cdot(-4)}}{2}$$

$$= \frac{3 \pm \sqrt{9 + 16}}{2}$$

$$= \frac{3 \pm 5}{2} = \frac{3+5}{2}, \frac{3-5}{2}$$

$$\therefore \lambda_1 = 4, \quad \lambda_2 = -1.$$

$$\lambda^2 - 3\lambda - 4 = 0.$$

$$\Rightarrow \lambda^2 - 4\lambda + \lambda - 4 = 0.$$

$$\Rightarrow \lambda(\lambda - 4) + 1(\lambda - 4) = 0.$$

$$\Rightarrow (\lambda - 4)(\lambda + 1) = 0.$$

$$\Rightarrow \lambda_1 = 4, \quad \lambda_2 = -1.$$

4

1 2 4

3 Eigenvectors

Note: Eigenvector corresponding to an eigenvalue of a matrix must be a non zero vector.

Example: 1 Let

$$A = \begin{pmatrix} 7 & 0 \\ -2 & 3 \end{pmatrix}.$$

From the previous computation, the eigenvalues are

$$\lambda_1 = 7, \quad \lambda_2 = 3.$$

We will find eigenvectors corresponding to each eigenvalue.

Let $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ be the eigenvector corresponding to the eigenvalue $\lambda_1 = 7$.

Then, $(A - \lambda_1)\vec{x} = \vec{0}$

$$\Rightarrow \begin{pmatrix} 7-\lambda_1 & 0 \\ -2 & 3-\lambda_1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 7-7 & 0 \\ -2 & 3-7 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & 0 \\ -2 & -4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

$$\Rightarrow \begin{pmatrix} 0\cdot x_1 + 0\cdot x_2 \\ -2x_1 - 4x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{array}{l} 0\cdot x_1 + 0\cdot x_2 = 0. \quad (1) \\ -2x_1 - 4x_2 = 0. \quad (2) \end{array}$$

$$x_2 = -\frac{1}{2}x_1$$

(2) $\Rightarrow -2x_1 = 4x_2$

$\Rightarrow x_1 = -2x_2$

Here x_2 is arbitrary any real number $(\neq 0)$.

Let $x_2 = 1$. $\therefore x_1 = -2$.

So, $\vec{x} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$.

Note for $c \neq 0, c \in \mathbb{R}$, $c\vec{x}$ is also an eigenvector corresponding to

For $\lambda_2 = 3$, let $\vec{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ be the eigenvector.

Then, $(A - \lambda_2 I)\vec{y} = \vec{0}$

$$\Rightarrow \begin{pmatrix} 7-3 & 0 \\ -2 & 3-3 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$ax = 0$
$a = 0$ or
$x = 0$
So if $a \neq 0$,
Then x must be zero.

$$\Rightarrow \begin{pmatrix} 4 & 0 \\ -2 & 0 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4y_1 \\ -2y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \left.\begin{array}{l} 4y_1 = 0 \\ -2y_1 = 0 \end{array}\right\} \Rightarrow y_1 = 0.$$

Here y_2 is arbitrary and let $y_2 = 1$.

$\therefore \vec{y} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.